

interconnected computer systems across the value and supply chain, these technologies are assisting in the digital transformation of manufacturing.

Adopting Industry 4.0, digital manufacturing, and the associated interconnectivity enables businesses to reap a variety of benefits, including improved operational performance and increased agility.

Embedded Internet of Things (EIoT) and Industry 4.0

Embedded Internet of Things (EIoT) refers to the integration of embedded devices (including sensors and open source hardware) with IoT based platforms and communication channels to enable real-time communication for varied applications. EIoT with Industry 4.0 is a technology shift of traditional engineering and governance to smart deployments. The three technical themes (connectivity, intelligence, and flexible automation) driving this transformation are collectively referred to as Industry 4.0, or the fourth industrial revolution with embedded devices.

The fourth industrial revolution represents breakthroughs in technologies, such as robots, intelligent systems, nanotechnology, quantum computing, microbiology, the Internet of Things, 3D printing, and autonomous cars.

In order to build a cyber-physical environment, Industry 4.0 combines OT (operational technology) and IT. A number of software platforms and libraries can be used for the simulation of EIoT and Industry 4.0 applications (Table 1).

Some use cases

In Industry 4.0 and EIoT simulations, there is a need to integrate lightweight protocols so that low-powered and smart gadgets can communicate effectively. MQTT is a key protocol that is used in Industry 4.0 and IoT based deployments.



Figure 1: Key segments of Industry 4.0

Table 1: Platforms and software suites for simulation of Industry 4.0 applications

Platform/Software suite	URL
Cloud MQTT	cloudmqtt.com
Mosquitto MQTT Broker	mosquitto.org
FlowHub	flowhub.io
Sumo	eclipse.org/sumo
NoFloJS	noflowjs.org
PraxisLive	praxislive.org
Netron	github.com/lutzroeder/netron
Ryven	gentle-sand-0114c7110.azurestaticapps.net/
PyFlow	wonderworks-software.github.io/PyFlow/
Wylodrin	wylodrin.com
Flutter Wireless	flutterwireless.com
SiteWhere	sitewhere.io
DSA	iot-dsa.org
Zetta	zettajs.org
Node Red	nodered.org
IoTivity	iotivity.org
Open IoT	openiot.eu